

Exponentials and Logarithms

Fast growth, and the function that undoes it · video 1:59

This lesson is about two functions that appear all over LLMs. The **exponential** e^x grows by multiplying: it is always positive, and it turns small differences into big ones. The **logarithm** $\ln x$ undoes it: it turns probabilities into negative numbers that are easy to compare, and it turns multiplication into addition.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 10).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. Growth by multiplying	▶ 0:08-0:26	4	☐
2. The number e, and e^x is always positive	▶ 0:26-0:46	4	☐
3. e^x exaggerates differences	▶ 0:47-1:00	4	☐
4. The logarithm undoes the exponential	▶ 1:00-1:13	4	☐
5. Logs of probabilities	▶ 1:13-1:28	4	☐
6. Multiplication becomes addition	▶ 1:29-1:40	3	☐
7. exp and log in NumPy	▶ 1:41-1:57	3	☐

Before you start: no earlier Foundations video is needed, only powers such as $2^3 = 2 \times 2 \times 2 = 8$. You will meet the exponential inside softmax (F07, and episodes 6 and 10), the logarithm in the training loss of episode 11, and a log-scale chart in episode 12.

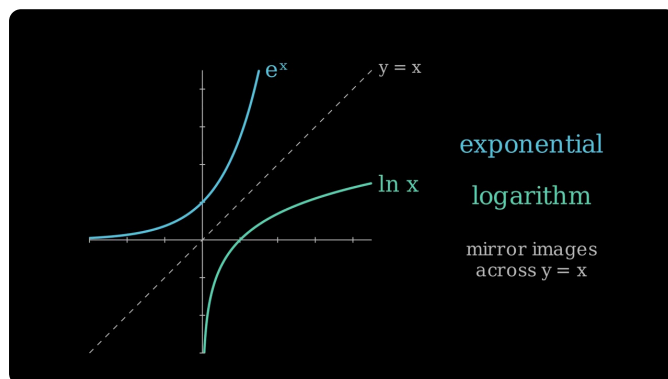
CONCEPT 1 OF 7

Growth by multiplying

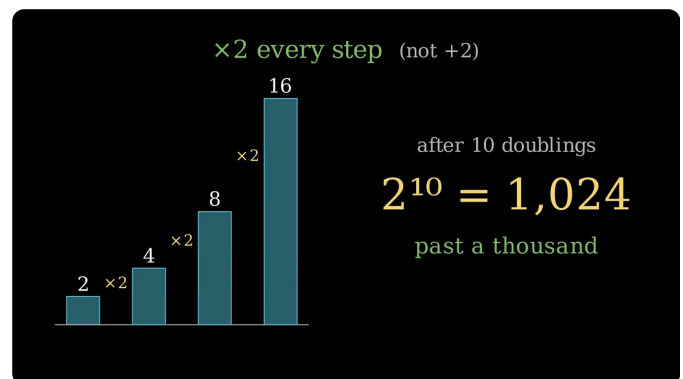
▶ Video 0:08-0:26

Two functions show up again and again in LLMs: the **exponential** and its opposite, the **logarithm**. Their graphs are mirror images of each other.

Exponential growth means **multiplying** by the same number at every step, not adding. Doubling gives 2, 4, 8, 16, ... and after just ten doublings you are past a thousand: $2^{10} = 1,024$. Adding 2 ten times would only get you to 20.



▶ 0:15 The exponential e^x and its opposite, the logarithm $\ln x$: mirror images across the line $y = x$.



▶ 0:25 Doubling every step: 2, 4, 8, 16, ... After ten doublings, $2^{10} = 1,024$.

KEY IDEA

Exponential = **repeated multiplication**. Every step multiplies by the same factor, so it soon outruns anything that adds: $2^{10} = 1,024$.

Check 1: Growth by multiplying

Answer all 4 questions before looking at the answers. Pass mark: all correct.

1.1 MULTIPLE CHOICE

Which sequence grows exponentially?

- A. 2, 4, 6, 8, 10
- B. 1, 4, 9, 16, 25
- C. 3, 6, 12, 24, 48
- D. 5, 10, 15, 20, 25

1.2 CALCULATION

Start from 1. (a) Double it 10 times. (b) Instead, add 2 ten times. (c) Double it 20 times.

1.3 CALCULATION

A number starts at 1 and is multiplied by 3 at every step. What is it after 5 steps?

1.4 TRUE OR FALSE

"In exponential growth, the amount added at each step stays the same."

True ☐ False ☐

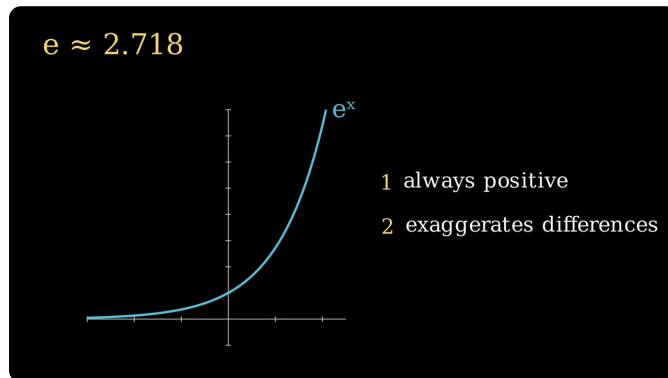
CONCEPT 2 OF 7

The number e , and e^x is always positive

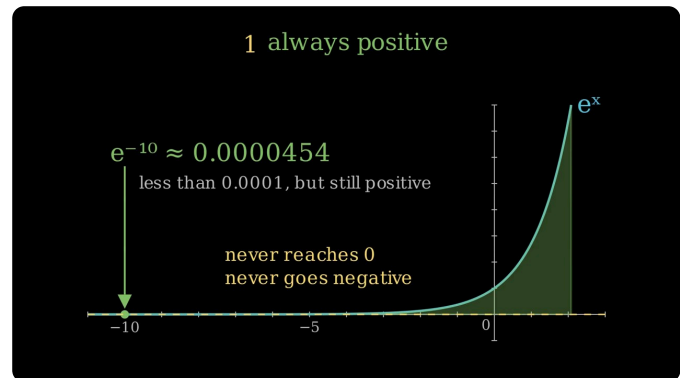
▶ Video 0:26–0:46

In machine learning the base is usually the number $e \approx 2.718$ (Euler's number), and the function is e^x , “e to the x”. Just like doubling, every +1 in x multiplies e^x by e: $e^0 = 1$, $e^1 \approx 2.718$, $e^2 \approx 7.39$, $e^3 \approx 20.1$.

Property 1: e^x is **always positive**. Negative inputs give small numbers, never negative ones: $e^{-1} \approx 0.368$, and $e^{-10} \approx 0.0000454$, less than one ten-thousandth. It never reaches zero.



▶ 0:35 The usual base is $e \approx 2.718$. The function e^x has two properties we care about.



▶ 0:46 Property 1: $e^{-10} \approx 0.0000454$, less than 0.0001 but still positive.

KEY IDEA

$e^x > 0$ for every x. Very negative inputs give tiny positive outputs, $e^0 = 1$, and each +1 in x multiplies the output by $e \approx 2.718$.

Check 2: The number e , and e^x is always positive

Answer all 4 questions before looking at the answers. Pass mark: all correct.

2.1 TRUE OR FALSE

“For a very negative input, such as $x = -100$, e^x is exactly zero.”

True ☐ False ☐

2.2 MULTIPLE CHOICE

Which list could be the outputs of e^x for three inputs?

- A. 1, 0.37, 7.39
- B. -1, 1, 2.72
- C. 0, 1, 2.72
- D. 2.72, -7.39, 20.1

2.3 CALCULATION

Using $e \approx 2.718$, find (a) e^0 , (b) e^2 and (c) $e^{-1} = 1/e$.

2.4 CALCULATION

The scores -2, 0 and 3 go through e^x . Roughly what comes out? Is any output negative?

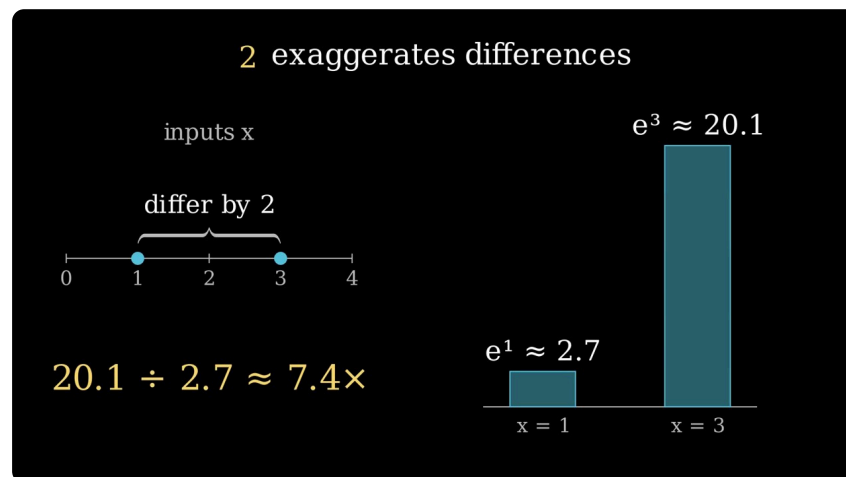
CONCEPT 3 OF 7

e^x exaggerates differences

▶ Video 0:47–1:00

Property 2: e^x **exaggerates differences**. The inputs 1 and 3 differ by just 2. But $e^1 \approx 2.7$ and $e^3 \approx 20.1$, so now one is **more than seven times** the other: $20.1 \div 2.7 \approx 7.4$ (exactly, $e^3/e^1 = e^2 \approx 7.39$).

The reason is concept 2: every +1 in the input multiplies the output by e . A gap of 2 in the inputs becomes a factor of $e^2 \approx 7.39$ in the outputs, wherever the two inputs are.



▶ 0:59 Inputs 1 and 3 differ by only 2, but $e^3 \approx 20.1$ is about 7.4 times $e^1 \approx 2.7$.

KEY IDEA

e^x turns **differences** into **ratios**: inputs that differ by d give outputs whose ratio is e^d . A small lead in the input becomes a big lead in the output.

Check 3: e^x exaggerates differences

Answer all 4 questions before looking at the answers. Pass mark: all correct.

3.1 CALCULATION

$e^1 \approx 2.718$ and $e^3 \approx 20.09$. Compute $e^3 \div e^1$ to two decimals. Which power of e is it?

3.2 CALCULATION

The inputs 10 and 12 also differ by 2. How many times bigger is e^{12} than e^{10} ?

3.3 MULTIPLE CHOICE

Two scores are 2 and 5. After e^x , how does the larger output compare with the smaller?

- A. 2.5 times bigger ($5 \div 2$)
- B. Exactly 3 bigger
- C. About 20 times bigger
- D. About the same

3.4 TRUE OR FALSE

" e^x keeps ratios: if one input is 3 times another, its output is also 3 times bigger."

True ☐ False ☐

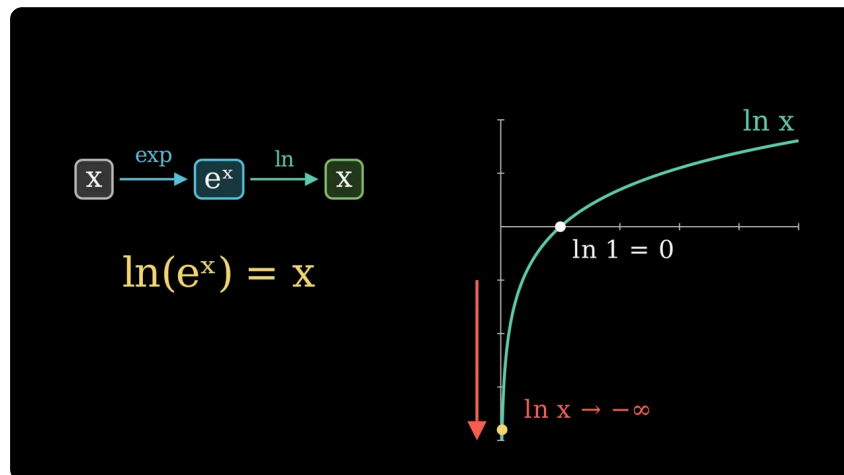
CONCEPT 4 OF 7

The logarithm undoes the exponential

▶ Video 1:00-1:13

The **natural logarithm**, $\ln$, undoes the exponential. It answers the question “e to *what* gives this number?”, so $\ln(e^x) = x$: go through exp and then ln, and you are back where you started. For example $\ln 20.1 \approx 3$, because $e^3 \approx 20.1$.

Two landmarks: $\ln 1 = 0$ (because $e^0 = 1$), and as the input shrinks toward zero, $\ln x$ **dives toward minus infinity**. So numbers between 0 and 1 have **negative** logs.



▶ 1:12 $\ln$ undoes exp: $\ln(e^x) = x$. $\ln 1 = 0$, and as x shrinks toward 0, $\ln x$ dives toward $-\infty$.

KEY IDEA

$\ln(e^x) = x$. $\ln 1 = 0$; $\ln x$ is negative for x between 0 and 1 and heads to $-\infty$ as x approaches 0; it is positive for $x > 1$.

Check 4: The logarithm undoes the exponential

Answer all 4 questions before looking at the answers. Pass mark: all correct.

4.1 CALCULATION

Without a calculator: (a) $\ln(e^5)$ (b) $\ln 1$ (c) $\ln e$ (d) $\ln 7.39$ (hint: concept 3).

4.2 PUT IN ORDER

Put in order from smallest to largest: $\ln 2 \cdot \ln 0.01 \cdot \ln 1 \cdot \ln 0.5$.

4.3 TRUE OR FALSE

“ $\ln 0.5$ is a positive number.”

True ☐ False ☐

4.4 MULTIPLE CHOICE

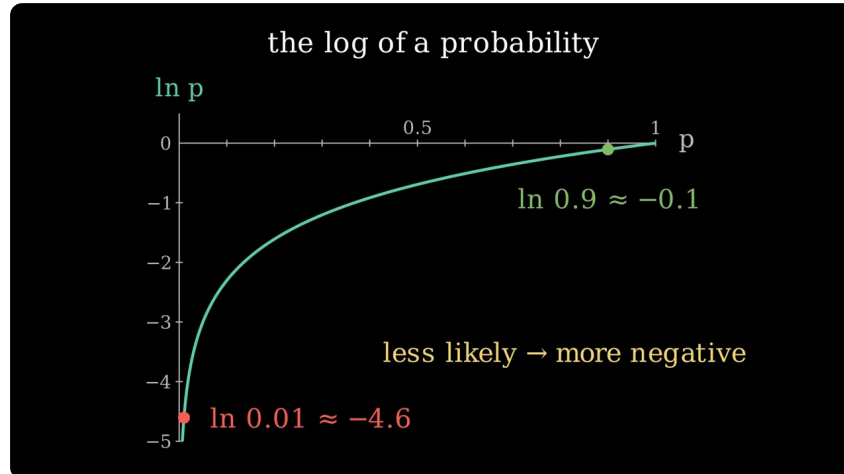
What happens to $\ln x$ as x gets closer and closer to 0 (0.1, 0.01, 0.001, ...)?

- A. It approaches 0
- B. It approaches 1
- C. It dives toward $-\infty$
- D. It becomes positive

Logs of probabilities

▶ Video 1:13–1:28

Probabilities live between 0 and 1, exactly where $\ln$ is **negative**. That makes logs perfect for probabilities. A likely event has a log **close to zero**: $\ln 0.9 \approx -0.105$. An unlikely one is strongly negative: $\ln 0.01 \approx -4.605$. A certain event, $p = 1$, has $\ln 1 = 0$.



▶ 1:28 $\ln p$ for probabilities: 0.9 gives about -0.1 , 0.01 gives about -4.6 . The less likely, the more negative.

KEY IDEA

For a probability p , $\ln p \leq 0$. **The less likely, the more negative**: likely events sit near 0 ($\ln 0.9 \approx -0.105$), unlikely ones far below ($\ln 0.01 \approx -4.605$).

Check 5: Logs of probabilities

Answer all 4 questions before looking at the answers. Pass mark: all correct.

5.1 MULTIPLE CHOICE

Model A gives the correct next word probability 0.9; model B gives it 0.01. Which statement is right?

- A. A's log-probability (≈ -0.105) is closer to 0, so A did better
- B. B's log-probability is closer to 0, so B did better
- C. Both log-probabilities are positive
- D. Logs can't compare probabilities

5.2 TRUE OR FALSE

"The log of a probability is never positive."

True ☐ False ☐

5.3 CALCULATION

Without a calculator, using the two landmarks above: (a) is $\ln 0.99$ closer to 0 or to -1 ? (b) Is $\ln 0.001$ above or below -4.605 ?

5.4 SHORT ANSWER

A model's log-probability for the correct word is -4.605 . What probability did it give that word? Was it confident?

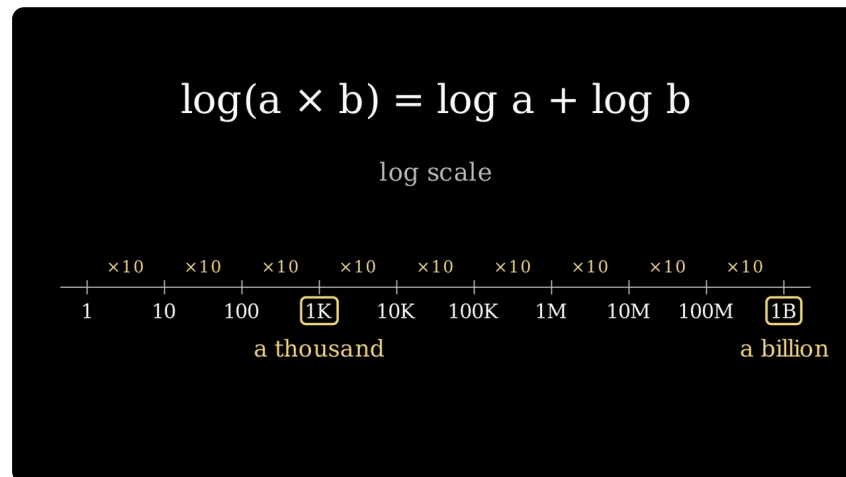
CONCEPT 6 OF 7

Multiplication becomes addition

▶ Video 1:29-1:40

Logs turn **multiplication into addition**: $\log(a \times b) = \log a + \log b$. For example, $\ln 6 + \ln 7 \approx 1.792 + 1.946 = 3.738 \approx \ln 42$. This matters for text: the probability of a whole sentence is the **product** of its words' probabilities (each given the words before it), and its log is simply the **sum** of the words' logs.

Logs also give **log-scale** charts: each tick is **ten times** the one before (1, 10, 100, ...), so a thousand and a billion fit on the same axis.



▶ 1:39 $\log(a \times b) = \log a + \log b$. On a log scale each step is $\times 10$, so a thousand and a billion fit on one axis.

KEY IDEA

$\log(a \times b) = \log a + \log b$: products of many probabilities become sums of logs. On a log scale, equal steps mean equal **factors** ($\times 10$).

Check 6: Multiplication becomes addition

Answer all 3 questions before looking at the answers. Pass mark: all correct.

6.1 CALCULATION

A three-word sentence has word probabilities 0.5, 0.2 and 0.1. (a) Multiply them. (b) Add their logs: $\ln 0.5 \approx -0.693$, $\ln 0.2 \approx -1.609$, $\ln 0.1 \approx -2.303$. (c) Compare (b) with $\ln 0.01$ from concept 5.

6.2 CALCULATION

$\ln 0.2 \approx -1.609$ and $\ln 0.1 \approx -2.303$. By how much does the log drop when a probability halves? Use it to predict $\ln 0.005$ from $\ln 0.01 \approx -4.605$.

6.3 CALCULATION

On the video's log-scale axis (1, 10, 100, 1K, ..., 1B), how many $\times 10$ steps is it from a thousand to a billion? Between which two ticks is 50,000?

CONCEPT 7 OF 7

exp and log in NumPy

▶ Video 1:41-1:57

In NumPy it's `np.exp` and `np.log` (`np.log` is the natural log, $\ln$). Both work on a single number or on every number of an array at once:

```
1 np.exp([1, 3])      # array([ 2.718, 20.086])
2 np.exp(-10)         # 4.54e-05
3 np.log(0.9)         # -0.105
4 np.log(0.01)        # -4.605
```

▶ 1:42 The code from the video: `np.exp` and `np.log`.

```
np.exp([1, 3])      # array([ 2.718, 20.086])
np.exp(-10)         # 4.54e-05
np.log(0.9)         # -0.105
np.log(0.01)        # -4.605
```

Where you'll see them: the exponential inside **softmax** (episodes 6 and 10), the logarithm in the **loss** of episode 11, and the **log-scale chart** of episode 12.

KEY IDEA

`np.exp` turns scores into positive, exaggerated numbers; `np.log` turns probabilities into negative numbers you can add.

Check 7: exp and log in NumPy

Answer all 3 questions before looking at the answers. Pass mark: all correct.

7.1 SHORT ANSWER

What does `np.log(np.exp(4.2))` return? Why?

7.2 MULTIPLE CHOICE

`p` is an array of the probabilities of a sentence's words. Which expression gives the log of the whole sentence's probability as a sum?

- A. `np.log(p).sum()`
- B. `np.exp(p).sum()`
- C. `np.log(p.sum())`
- D. `np.log(p).prod()`

7.3 HANDS-ON CODE

Try it yourself. Predict each printed line before running the code. (a) What do the first two lines print (2 decimals)? Which number from concept 3 is the second one? (b) What does the third line print, and why? (c) The fourth line prints two numbers: which, and how are they related? (d) `q.prod()` prints `0.0`. Is the true product zero? What does the sum of logs give instead?

```
import numpy as np

x = np.array([1.0, 2.0, 3.0])
print(np.exp(x))
print(np.exp(x[2]) / np.exp(x[0]))
print(np.log(np.exp(x)))

p = np.array([0.5, 0.2, 0.1])      # probabilities of 3 words
print(p.prod(), np.log(p).sum())

q = np.full(1000, 0.01)           # 1,000 words, each 0.01
print(q.prod(), np.log(q).sum())
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. Growth by multiplying · rewatch ▶ 0:08-0:26

1.1 C. Each term is the previous one $\times 2$. A and D add a fixed amount; B are squares ($1^2, 2^2, 3^2, \dots$), whose step-to-step factor keeps shrinking.

1.2 (a) 1,024 (b) 21 (c) 1,048,576. $2^{20} = 2^{10} \times 2^{10} = 1,024 \times 1,024$: past a million, while adding only reaches 21.

1.3 243. $3^5 = 3 \times 3 \times 3 \times 3 \times 3 = 243$. Same idea as doubling, with factor 3.

1.4 False. The *factor* stays the same ($\times 2$); the amount added keeps growing: +2, +4, +8, +16, ...

2. The number e, and e^x is always positive · rewatch ▶ 0:26-0:46

2.1 False. It is extremely small, but still positive. e^x never reaches zero.

2.2 A. These are e^0, e^{-1} and e^2 . The others contain 0 or a negative number, which e^x never produces.

2.3 (a) 1 (b) ≈ 7.39 (c) ≈ 0.368 . $e^2 = 2.718 \times 2.718 \approx 7.39$; $e^{-1} = 1 \div 2.718 \approx 0.368$; zero steps of multiplying leaves 1.

2.4 ≈ 0.135 , 1 and 20.1. None is negative. Even the negative score gives a positive number. This is what lets softmax (F07) turn any scores into positive numbers.

3. e^x exaggerates differences · rewatch ▶ 0:47-1:00

3.1 $\approx 7.39 = e^2$. $20.09 \div 2.718 \approx 7.39$: two extra factors of e.

3.2 ≈ 7.39 times, the same as before. Only the difference matters: $e^{12} \div e^{10} = e^2 \approx 7.39$.

3.3 C. The inputs differ by 3, so the outputs differ by a factor $e^3 \approx 20.1$.

3.4 False. Inputs 1 and 3 (ratio 3) give 2.7 and 20.1 (ratio ≈ 7.4). e^x exaggerates.

4. The logarithm undoes the exponential · rewatch ▶ 1:00-1:13

4.1 (a) 5 (b) 0 (c) 1 (d) ≈ 2 . In asks “e to what?”: e^5, e^0, e^1 and $e^2 \approx 7.39$.

4.2 $\ln 0.01 \rightarrow \ln 0.5 \rightarrow \ln 1 \rightarrow \ln 2$. About $-4.61, -0.69, 0$ and 0.69 . Bigger input, bigger log.

4.3 False. $\ln 0.5 \approx -0.693$. Every number between 0 and 1 has a negative log.

4.4 C. $\ln 0.1 \approx -2.3, \ln 0.01 \approx -4.6, \ln 0.001 \approx -6.9, \dots$ with no lower limit.

5. Logs of probabilities · rewatch ▶ 1:13-1:28

5.1 A. $\ln 0.9 \approx -0.105$ is near 0; $\ln 0.01 \approx -4.605$ is far below. Closer to 0 = more likely.

5.2 True. A probability is at most 1, and $\ln 1 = 0$; anything below 1 has a negative log.

5.3 (a) Very close to 0 (≈ -0.01). (b) Below (≈ -6.9). 0.99 is even more likely than 0.9, so its log is even nearer 0. 0.001 is less likely than 0.01, so its log is more negative.

5.4 0.01: not confident at all. $\ln 0.01 \approx -4.605$, so $p = e^{-4.605} \approx 0.01$. It thought the correct word was very unlikely.

6. Multiplication becomes addition · rewatch ► 1:29-1:40

6.1 (a) 0.01 (b) -4.605 (c) the same. The sum of the logs is the log of the product: $\ln 0.01 \approx -4.605$.

6.2 By ≈ 0.693 each time; $\ln 0.005 \approx -5.298$. Halving is $\times 0.5$, which adds $\ln 0.5 \approx -0.693$, wherever you start.

6.3 6 steps; between 10K and 100K. $1K \rightarrow 10K \rightarrow 100K \rightarrow 1M \rightarrow 10M \rightarrow 100M \rightarrow 1B$. 50,000 is more than 10,000 and less than 100,000.

7. exp and log in NumPy · rewatch ► 1:41-1:57

7.1 4.2. $\ln$ undoes $\exp$: $\ln(e^x) = x$.

7.2 A. The log of a product is the sum of the logs. C takes the log of a sum (wrong rule); D multiplies the logs.

7.3 (a) [2.72, 7.39, 20.09] and 7.39 · (b) [1. 2. 3.] · (c) 0.01 and -4.605 · (d) -4605.17 (a) e^1, e^2, e^3 ; the ratio $e^3/e^1 = e^2 \approx 7.39$ is the one from concept 3. (b) $\ln$ undoes $\exp$, so you get x back. (c) The product 0.01 (printed as 0.010000000000000002, a tiny rounding error) and the sum of logs $-4.605 = \ln 0.01$: the log of a product is the sum of the logs. (d) Not zero: 0.01 multiplied 1,000 times is 10^{-2000} , far too small for the computer to store, so it rounds to 0.0. The sum of logs, $1,000 \times (-4.605) \approx -4605.17$, is a perfectly ordinary number. This is one reason models work with log-probabilities.